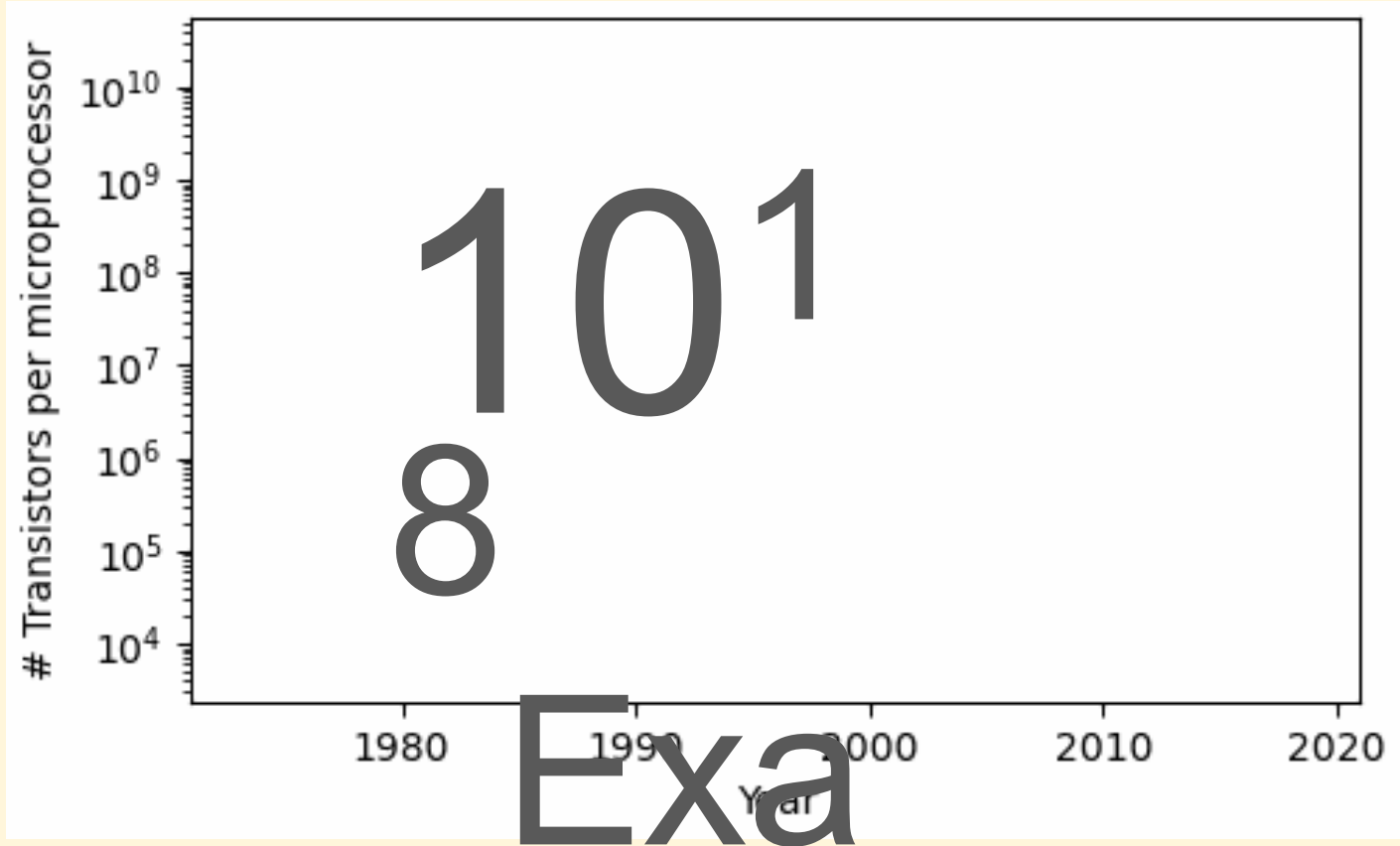


Assessing the quality of zarr datasets from km scale ESM simulations

Kameswar Rao Modali¹, Karsten Peters von Gehlen¹, Fabian Wachsmann¹,
Janos Zimmermann¹, Dogus Kaan Bilir¹, Florian Ziemen¹, Karl-Hermann
Wieners², Lukas Kluft², Siddhant Tibrewal²

2nd Workshop on Correctness and Reproducibility for Earth System Software
NCAR and virtual, 6-7 November 2025

Moore's Law



Source: OurWorldInData.org, Karl Rupp, Microprocessor Trend Data (2022)

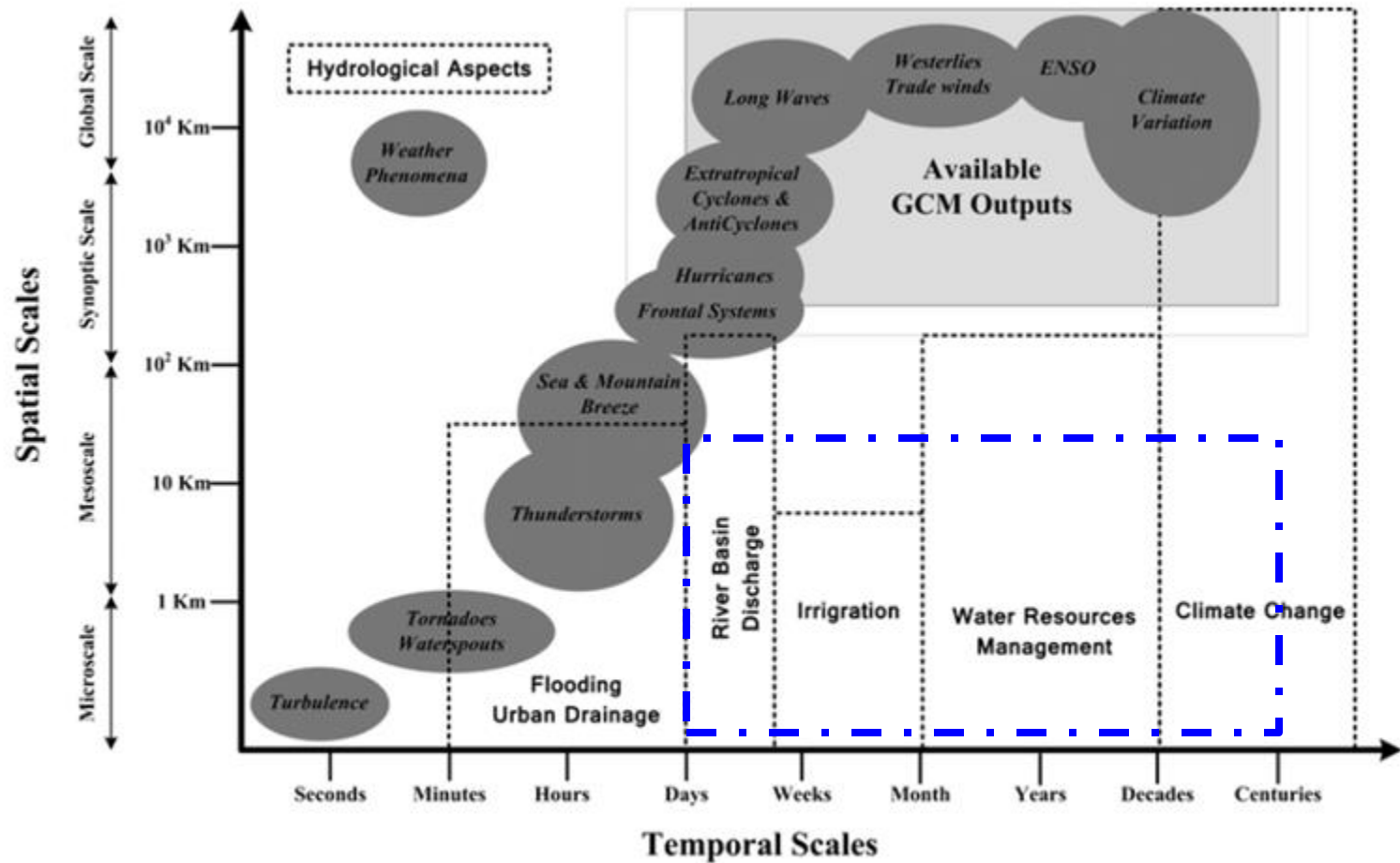
What do we expect from ESMs capable of leveraging Exascale computing power?

- Some parameterizations become obsolete -> more “realism”

What kind of simulations can be targeted with such ESMs?

- Possibility to compute and evaluate kilometer-scale climate trajectories.



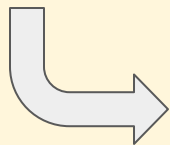


The spatial and temporal scale of weather phenomena, available GCMs outputs and hydrological impact evaluation (Adapted from Nese and Greci (2011))

Of course, new approaches come with challenges...

How do we cater to the gamut of length and time scales of scientific pursuits?

How do we enable efficient data access, analysis and evaluation?



Combination of HEALPix grid and
zarr-formatted data

Hierarchical equal area iso-latitude pixelization - HEALPix [\(Gorski et al. 2004\)](#)

- Hierarchical tessellation of sphere into curvilinear quadrilaterals - lowest resolution partition has 12 base pixels
- Resolution of the tessellation increases by division of each pixel into four new ones.
- Areas of all pixels at a given resolution are identical.
- Pixels are distributed on lines of constant latitude.

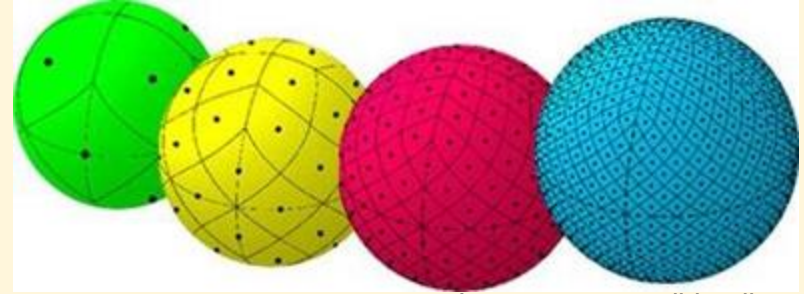


Image source: wikipedia

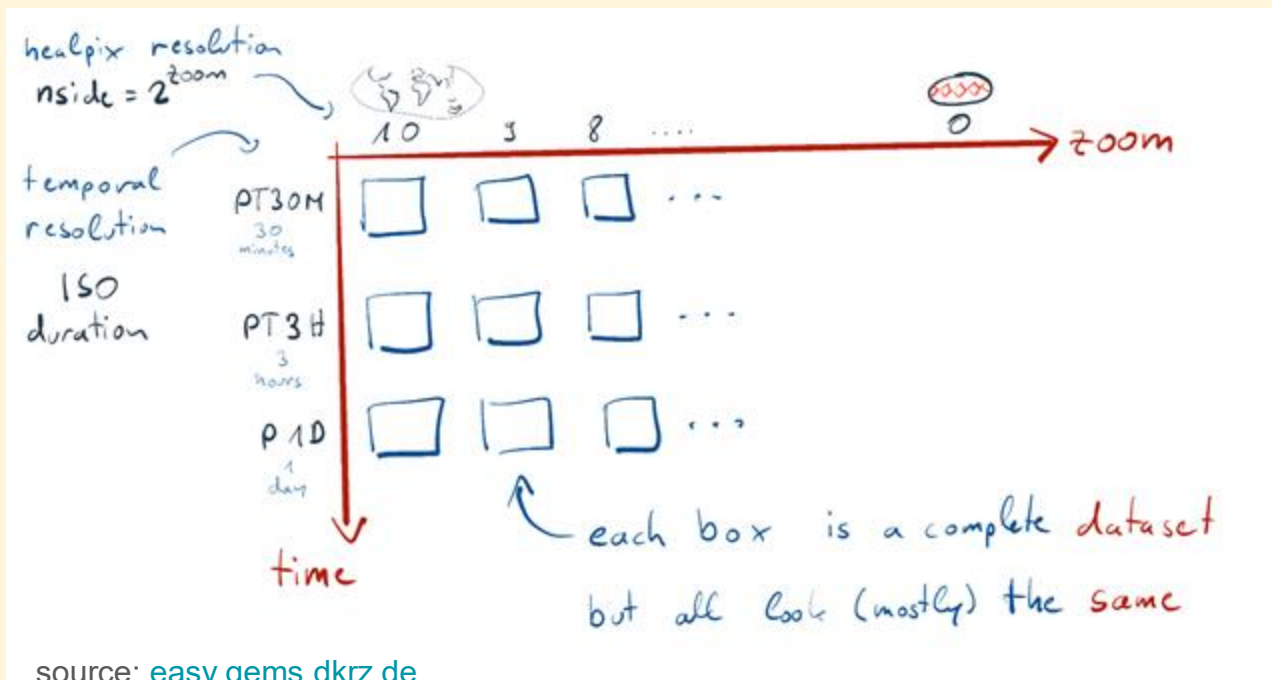
Mapping km-scale model output to HEALPix enables **efficient** analysis and evaluation of global quantities as well as feature analysis at maximum resolution.

HR ESM data on HEALPix, stored in zarr-format



Zarr

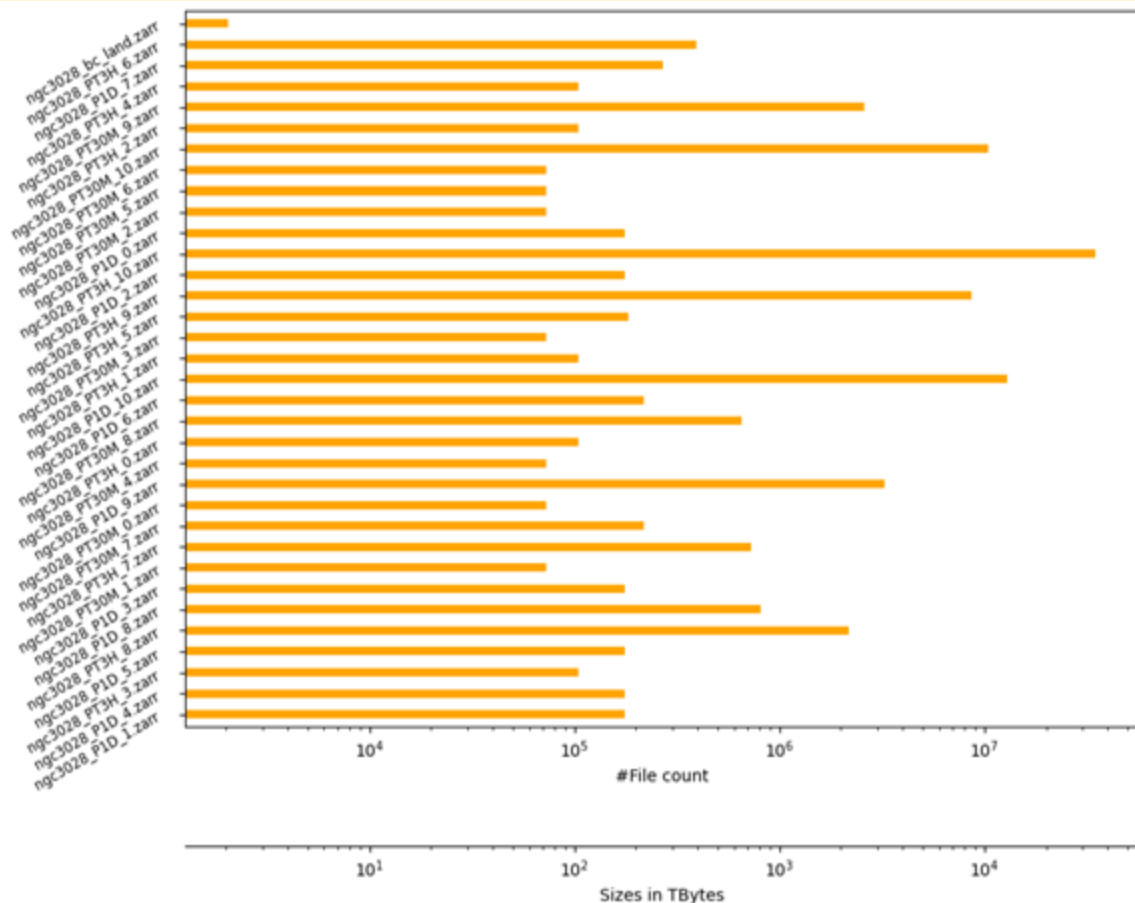
The output is stored as combination of multiple resolutions and chunks; example from the ICON model:



<https://zarr.readthedocs.io/en/stable/>

- one dataset per zoom level
 - (0-10)
- and temporal output
 - 30 minutes
 - 3 hours
 - 1 day

How does it work in the context of ICON model output?



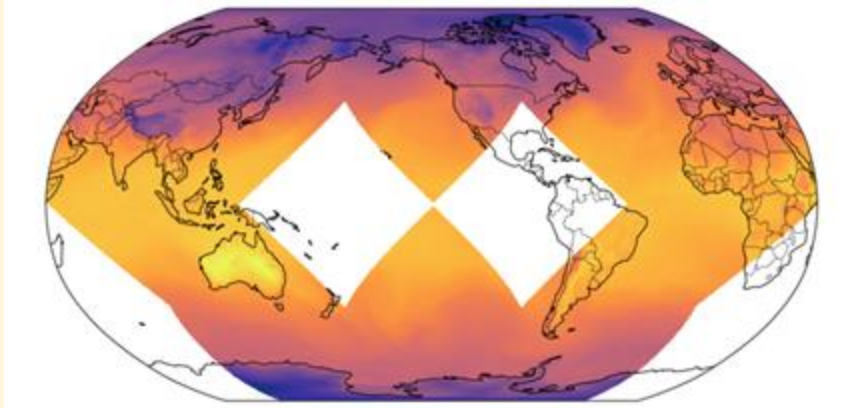
- up to 10^7 files/chunks per dataset
- up to hundreds of Tb per dataset

Technical challenge!!

How to ascertain the data integrity?

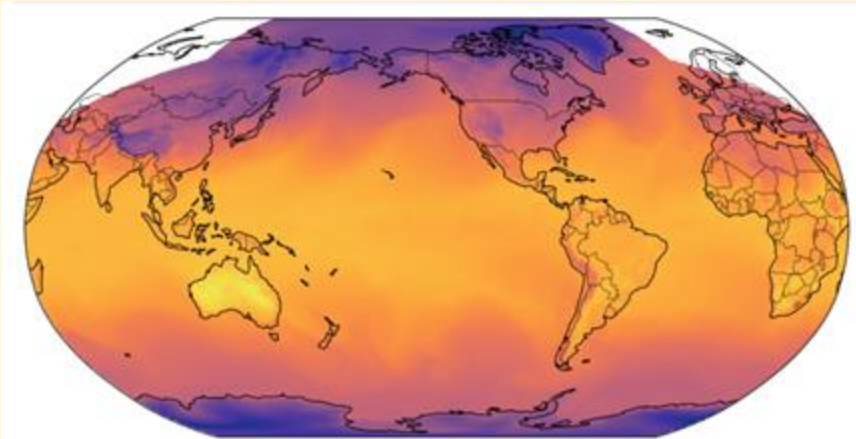
zoom 7

Single chunk
missing



zoom 9

Single chunk
missing



- Data is produced and moved across storage tiers (HDD -> S3 -> tape) which could result in missing chunks
- Effects could be
 - wrong chunk count,
 - uninitialized chunks,
 - missing chunks Vs zoom levels

How to ascertain the data integrity?

Typical Zarr store

atmos_fluxes_FrshFlux_Evaporation	dzghalf	ice_v	qv2m	tauv																									
atmos_fluxes_FrshFlux_Precipitation	FrshFlux_IceSalt	level_full	rlds	time																									
atmos_fluxes_FrshFlux_Runoff	FrshFlux_TotalIce	level_half	rlus	tke																									
atmos_fluxes_FrshFlux_SnowFall	heat_content_seaice	mlotst	rlut	to																									
atmos_fluxes_HeatFlux_Latent	heat_content_snow	newice	rsds	ts																									
atmos_fluxes_HeatFlux_LongWave	heat_content_total	ocean_fraction_depth_full	rsdt	u																									
atmos_fluxes_HeatFlux_Sensible	heatOceI	ocean_fraction_depth_half	rsus	ua																									
atmos_fluxes_HeatFlux_ShortWave	heatOceW	ocean_fraction_surface	rsut	uas																									
atmos_fluxes_HeatFlux_Total	hfIs	pfull	seb_forc_hflx_box	u_vint																									
atmos_fluxes_stress_x	hfss	phalf	seb_heat_cap_box	v																									
atmos_fluxes_stress_xw	hI	or	sfcwind	va																									
atmos_fluxes_stres	0	11	121	133	145	157	169	180	192	203	215	227	239	250	262	274	286	298	309	320	332	344	356	39	50	62	74	86	98
atmos_fluxes_stres	1	110	122	134	146	158	17	181	193	204	216	228	24	251	263	275	287	299	31	321	333	345	357	4	51	63	75	87	99
A_tracer_v_to	10	111	123	135	147	159	170	182	194	205	217	229	240	252	264	276	288	3	310	322	334	346	358	40	52	64	76	88	
cli	100	112	124	136	148	16	171	183	195	206	218	23	241	253	265	277	289	30	311	323	335	347	359	41	53	65	77	89	
clivi	101	113	125	137	149	160	172	184	196	207	219	230	242	254	266	278	29	300	312	324	336	348	36	42	54	66	78	9	
clw	102	114	126	138	15	161	173	185	197	208	22	231	243	255	267	279	290	301	313	325	337	349	360	43	55	67	79	90	
conc	103	115	127	139	150	162	174	186	198	209	220	232	244	256	268	28	291	302	314	326	338	35	361	44	56	68	8	91	
crs	104	116	128	14	151	163	175	187	199	21	221	233	245	257	269	280	292	303	315	327	339	350	362	45	57	69	80	92	
delhi	105	117	129	140	152	164	176	188	2	210	222	234	246	258	27	281	293	304	316	328	34	351	363	46	58	7	81	93	
depth_full	106	118	13	141	153	165	177	189	20	211	223	235	247	259	270	282	294	305	317	329	340	352	364	47	59	70	82	94	
depth_half	107	119	130	142	154	166	178	19	200	212	224	236	248	26	271	283	295	306	318	33	341	353	365	48	6	71	83	95	
	108	12	131	143	155	167	179	190	201	213	225	237	249	260	272	284	296	307	319	330	342	354	37	49	60	72	84	96	
	109	120	132	144	156	168	18	191	202	214	226	238	25	261	273	285	297	308	32	331	343	355	38	5	61	73	85	97	

The name
nu

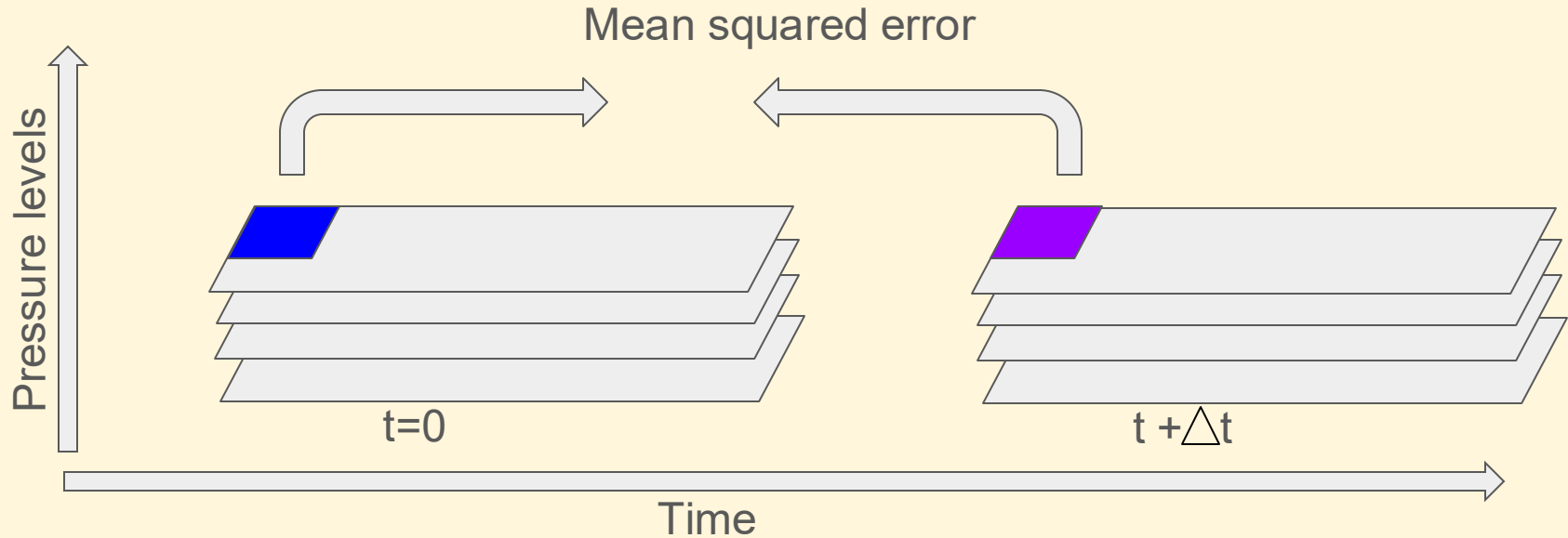
The names of chunks are numbers!!!

0	104	110	117	123	13	136	142	149	155	161	168	174	180	187	20	27	33	4	46	52	59	65	71	78	84	90	97
1	105	111	118	124	130	137	143	15	156	162	169	175	181	188	21	28	34	40	47	53	6	66	72	79	85	91	98
10	106	112	119	125	131	138	144	150	157	163	17	176	182	189	22	29	35	41	48	54	60	67	73	8	86	92	99
100	107	113	12	126	132	139	145	151	158	164	170	177	183	19	23	3	36	42	49	55	61	68	74	80	87	93	
101	108	114	120	127	133	14	146	152	159	165	171	178	184	190	24	30	37	43	5	56	62	69	75	81	88	94	
102	109	115	121	128	134	140	147	153	16	166	172	179	185	191	25	31	38	44	50	57	63	7	76	82	89	95	
103	11	116	122	129	135	141	148	154	160	167	173	18	186	2	26	32	39	45	51	58	64	70	77	83	9	96	

How to ascertain the data quality?

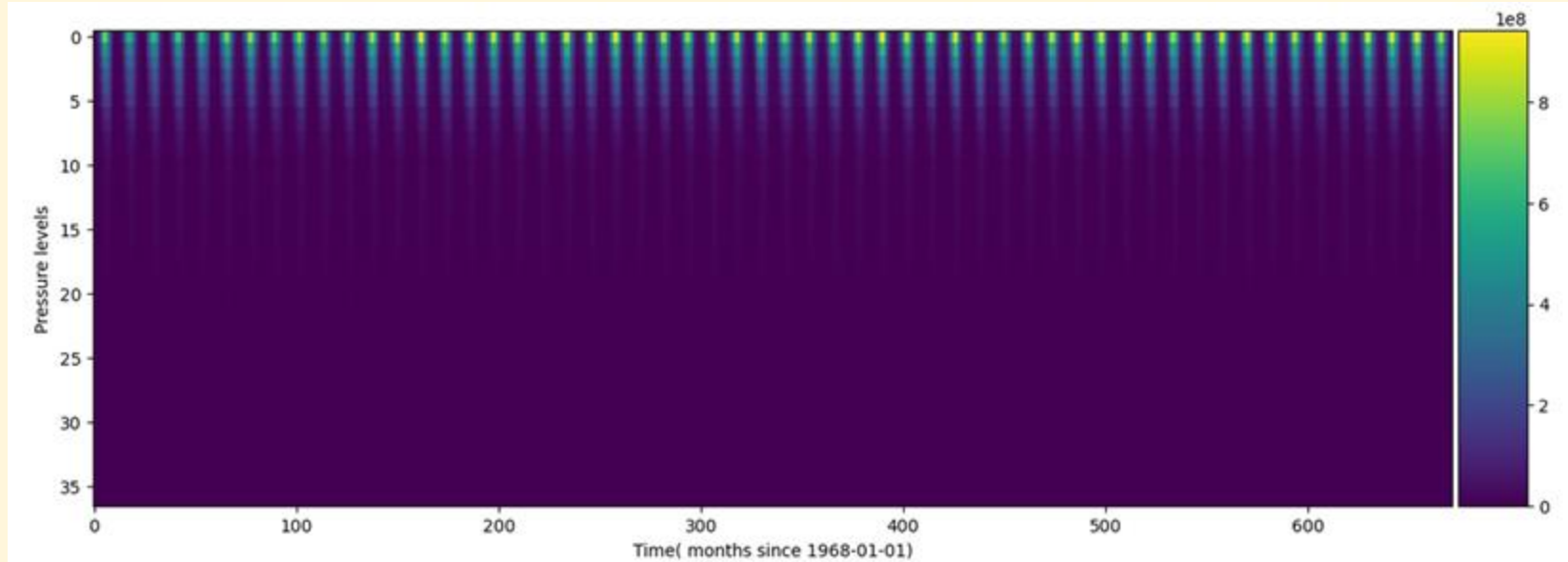
Mean Squared Error

$$MSE(X, Y) = \frac{1}{N} \sum_{i=1}^N (X_i - Y_i)^2$$



How to ascertain the data quality?

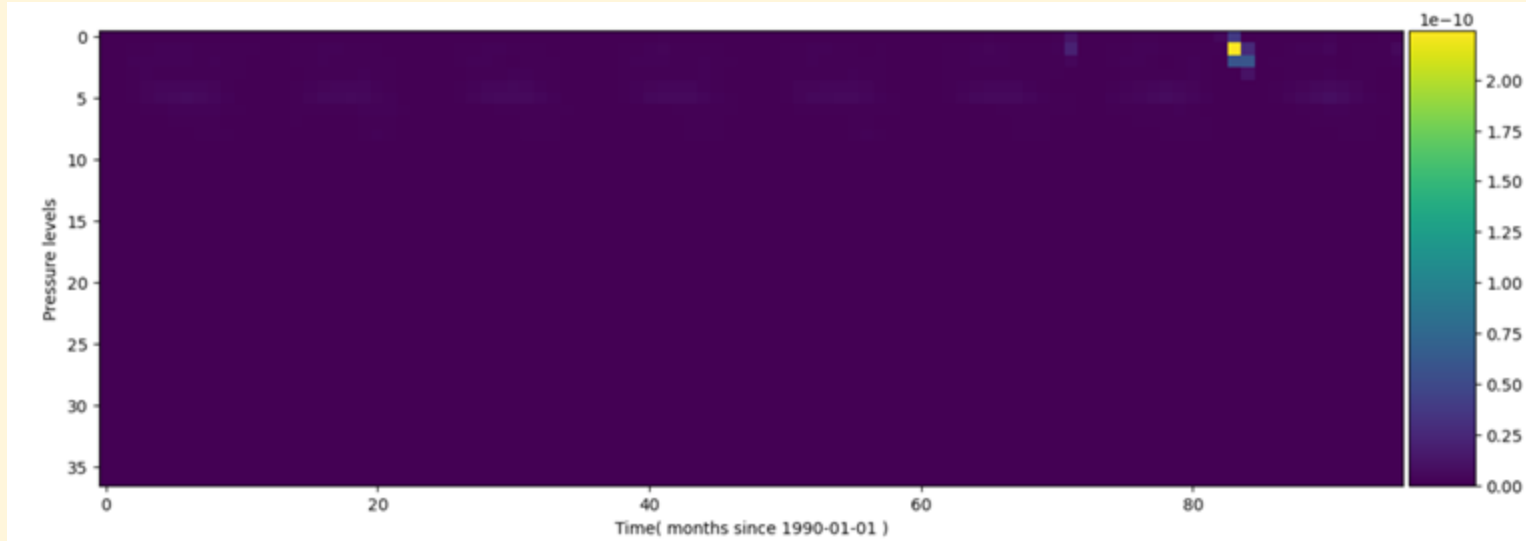
Mean Squared Error (MSE)



ERA5 Geopotential(mon) 1968-2020 250 Km² 37 pressure levels

How to ascertain the data quality?

Mean Squared Error (MSE)



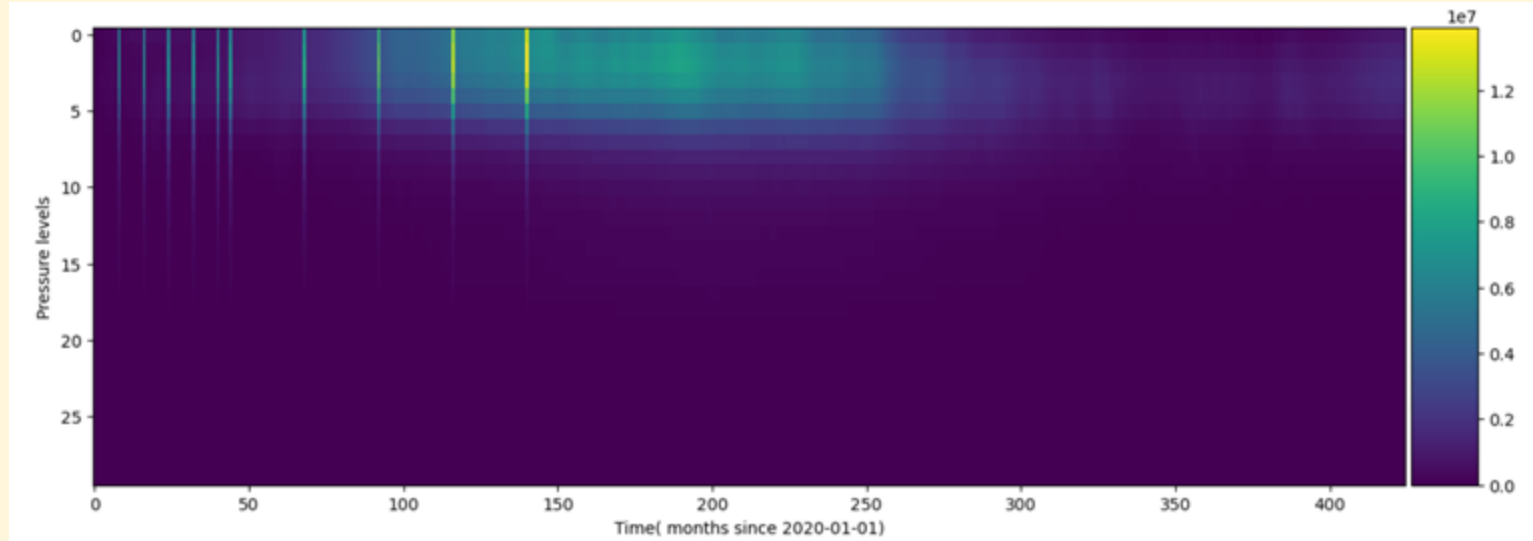
ERA5 Ozone mixing (mon) 1990-1997 250 Km² 37 pressure levels

It turns out that some issue with measuring instrument(TOMS) in 1995...

<https://www.theozonehole.org/ozone1995.htm>

How to ascertain the data quality?

Mean Squared Error (MSE)



ICON Geopotential(P1D)

It turns out that those abnormal streaks are due to some issue in computing daily values from model time step...

How to ascertain the data quality?

Structural Similarity Index Measure (SSIM)

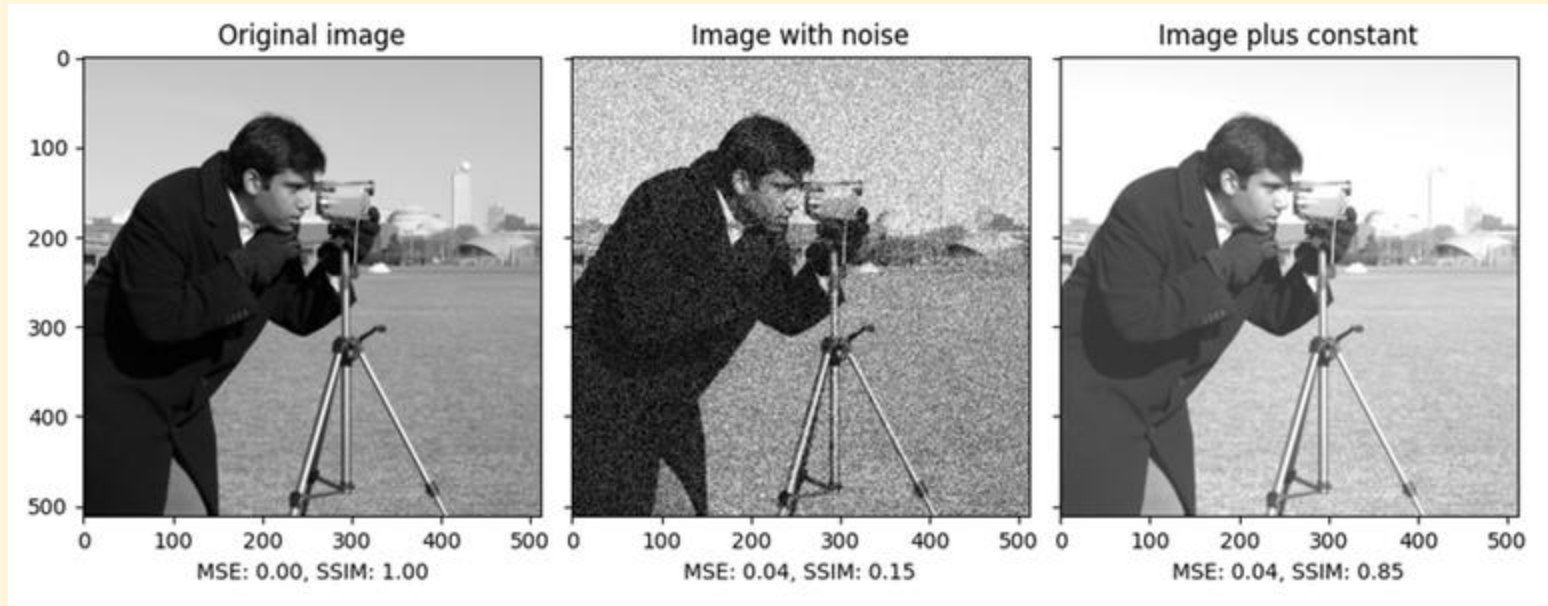
= $f(\textit{luminance}, \textit{contrast}, \textit{structure})$

$$SSIM(x_i, y_i) = (l(x_i, y_i))^\alpha \cdot (c(x_i, y_i))^\beta \cdot (s(x_i, y_i))^\gamma$$

$$l(x_i, y_i) = \frac{2\mu_{x_i}\mu_{y_i} + c_1}{\mu_{x_i}^2 + \mu_{y_i}^2 + c_1} \quad c(x_i, y_i) = \frac{2\sigma_{x_i}\sigma_{y_i} + c_2}{\sigma_{x_i}^2 + \sigma_{y_i}^2 + c_2} \quad s(x_i, y_i) = \frac{\sigma_{x_i y_i} + c_3}{\sigma_{x_i}\sigma_{y_i} + c_3}$$

How to ascertain the data quality?

Structural Similarity Index Measure (SSIM)



Source: <https://scikit-image.org/>

Evaluation of entire images for overall similarity -> global ESM simulation fields?

However, difficult to realize for high-resolution climate simulation data (too many pixels)

How to ascertain the data quality?

Data Structural Similarity Index Measure (DSSIM) ([Baker et al., 2024](#))

IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS, VOL. 30, NO. 9, SEPTEMBER 2024

6261

On a Structural Similarity Index Approach for Floating-Point Data

Allison H. Baker , Alexander Pinard , and Dorit M. Hammerling 

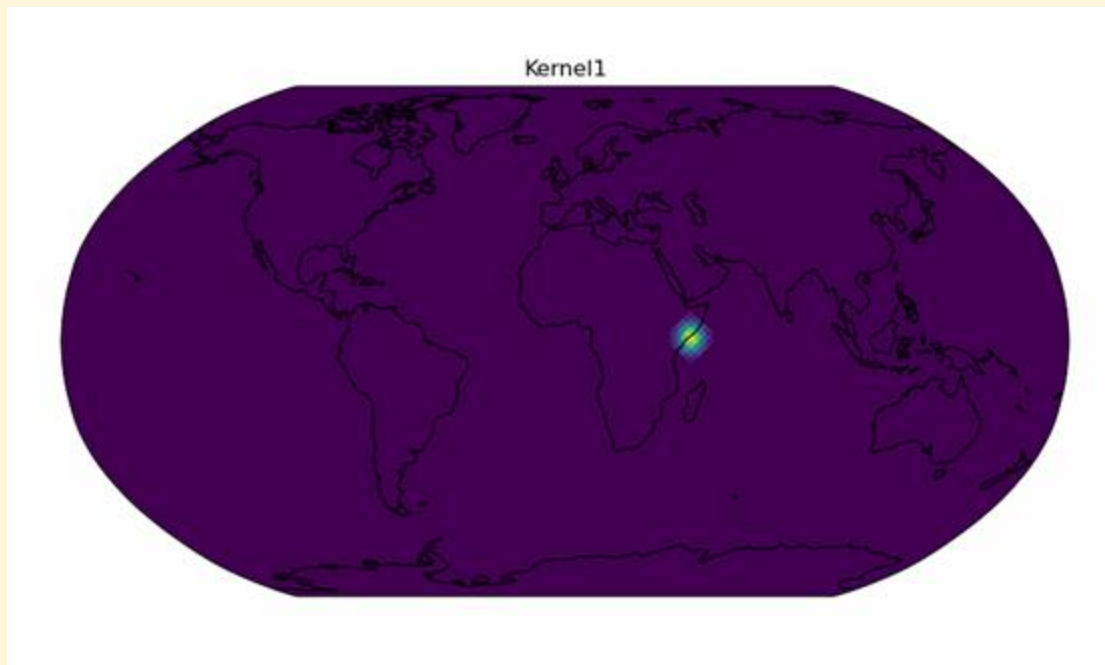
$$dssim(x_i, y_i) \leftarrow S_1(x_i, y_i) S_2(x_i, y_i) \quad S_1 = \frac{2\mu_{x_i} \mu_{y_i} + C_1}{\mu_{x_i}^2 + \mu_{y_i}^2 + C_1} \quad S_2 = \frac{2\sigma_{x_i} \sigma_{y_i} + C_2}{\sigma_{x_i}^2 + \sigma_{y_i}^2 + C_2}$$
$$DSSIM(X, Y) = \frac{1}{N} \sum_{i=1}^N dssim(x_i, y_i)$$

Applicable to regular grids; HR model output is often unstructured, e.g. HEALPix

How to ascertain the data quality?

Data Structural Similarity Index Measure (DSSIM)

Animation of kernel for
various grid cell
locations

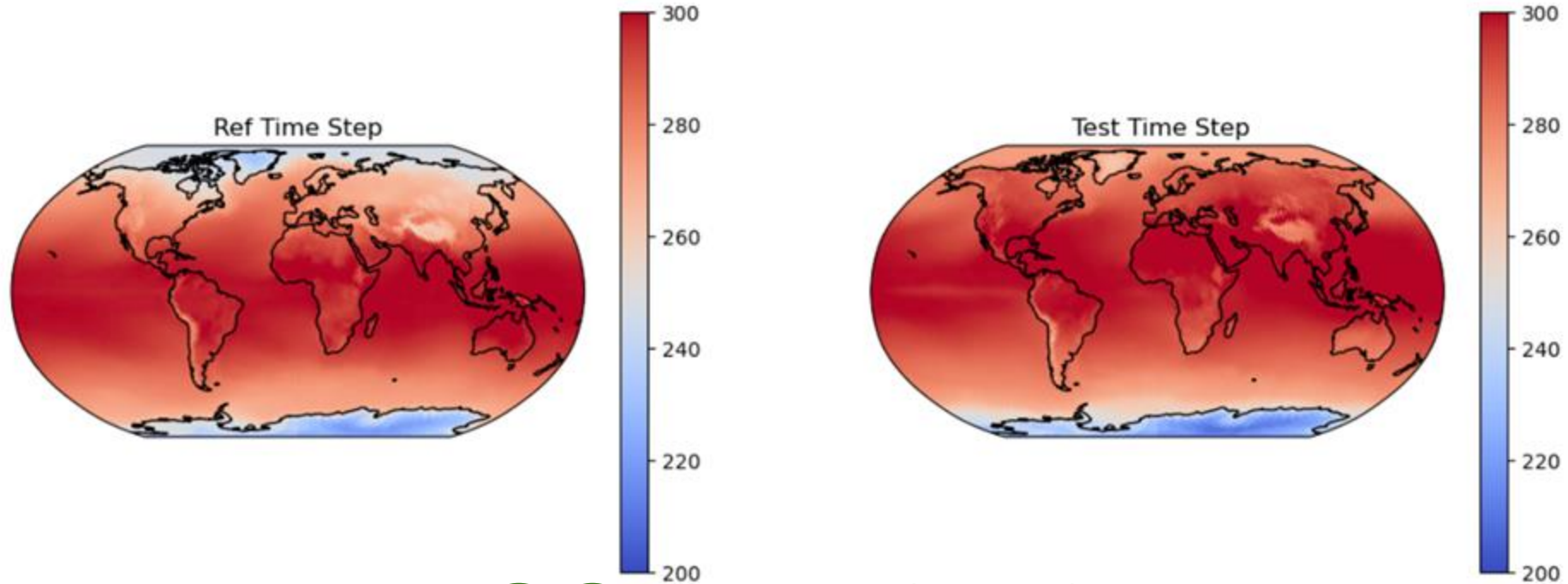


Adapted to HEALPix grid using 'healpix-convolution' package
for constructing the gaussian kernel

<https://healpix-convolution.readthedocs.io/en/latest/index.html>

How to ascertain the data quality?

Data Structural Similarity Index Measure (DSSIM)



!!!DSSIM = 0.56!!!

Summary

- Km-scale global (physical) ESM simulation data require **novel data layout and data formats** to allow processing
- Combination of cloud native formats like **Zarr** (in small chunks) and **HEALPix** require adaptation of techniques to enable ensuring data integrity and data quality .
- **Data integrity** (erroneous chunks) can be verified using **simple method**
- **Adapted DSSIM to HEALPix grid** which enables to assess data quality almost at the level of visual inspection.

Outlook

- Publication of code for community reuse
- Possibility to **incorporate into HR ESM workflows**

THANK YOU